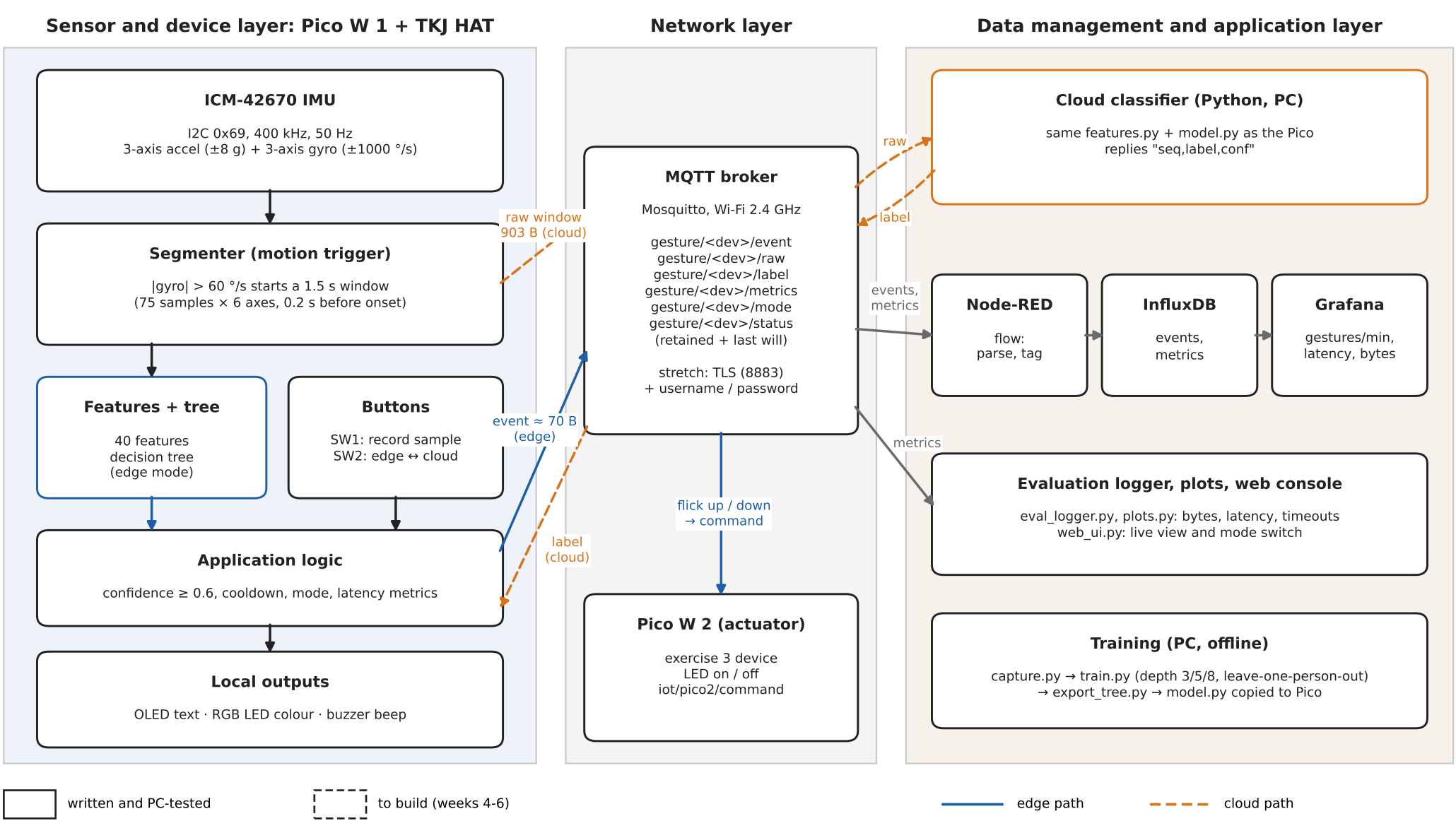
Sensor and device layer: Pico W 1 + TKJ HAT
Network layer
Data management and application layer
ICM-42670 IMU
I2C 0x69, 400 kHz, 50 Hz
3-axis accel (±8 g) + 3-axis gyro (±1000 °/s)
Segmenter (motion trigger)
|gyro| > 60 °/s starts a 1.5 s window
(75 samples × 6 axes, 0.2 s before onset)
Features + tree
40 features
decision tree
(edge mode)
Buttons
SW1: record sample
SW2: edge ↔ cloud
Application logic
confidence ≥ 0.6, cooldown, mode, latency metrics
Local outputs
OLED text · RGB LED colour · buzzer beep
raw window
903 B (cloud)
event ≈ 70 B
(edge)
label
(cloud)
MQTT broker
Mosquitto, Wi-Fi 2.4 GHz
gesture/<dev>/event
gesture/<dev>/raw
gesture/<dev>/label
gesture/<dev>/metrics
gesture/<dev>/mode
gesture/<dev>/status
(retained + last will)
stretch: TLS (8883)
+ username / password
flick up / down
→ command
Pico W 2 (actuator)
exercise 3 device
LED on / off
iot/pico2/command
raw
label
Cloud classifier (Python, PC)
same features.py + model.py as the Pico
replies "seq,label,conf"
events,
metrics
Node-RED
flow:
parse, tag
InfluxDB
events,
metrics
Grafana
gestures/min,
latency, bytes
metrics
Evaluation logger, plots, web console
eval_logger.py, plots.py: bytes, latency, timeouts
web_ui.py: live view and mode switch
Training (PC, offline)
capture.py → train.py (depth 3/5/8, leave-one-person-out)
→ export_tree.py → model.py copied to Pico
written and PC-tested
to build (weeks 4-6)
edge path
cloud path